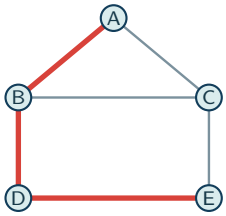
GRAPH THEORY LADDER 5 OF 9

Hamilton cycles

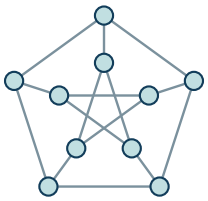
A cycle through every vertex once has no simple test, and the candidate tours explode in number.

1 / THE RULE AND THE PICTURE

A Hamilton cycle visits every vertex exactly once and returns to the start



one Hamilton cycle



Petersen graph, none

Hamilton devised a puzzle in 1857 asking for such a cycle on the dodecahedron, and there is no degree test that decides the question in general; two graphs of the same size can look alike and differ in having one:

Hamilton cycle: n vertices, n edges, no repeats

The house graph has exactly one, and the Petersen graph, drawn on the right, has none.

2 / EULER AND HAMILTON COMPARED

The two questions sound alike and behave very differently

	Euler	Hamilton
visits	every edge once (Euler)	every vertex once (Hamilton)
test	degrees all even: easy	no simple test known
house graph	a trail from B to C	the cycle A, B, D, E, C
hard version	none	the travelling salesman problem

3 / SUFFICIENT CONDITIONS

Dirac and Ore guarantee a cycle once degrees are large

Dirac's theorem says that a simple graph on $n \geq 3$ vertices in which every degree is at least $n/2$ has a Hamilton cycle, and Ore's theorem weakens this to $\deg u + \deg v \geq n$ for every non-adjacent pair.
For $K_{3,3}$ every degree is $3 \geq 3$, so a cycle exists, and indeed there are 6. The house graph has minimum degree $2 < 2.5$, so Dirac says nothing, yet it has a Hamilton cycle, which shows the conditions are sufficient and not necessary.

4 / A GRAPH WITHOUT ONE

The Petersen graph is the classic obstruction

The Petersen graph has 10 vertices, every degree 3, which is large enough to look promising, and an exhaustive search over all orderings shows that it has no Hamilton cycle, although it has Hamilton paths.
On the cube graph Q_3 there are 6 cycles, one of which is the Gray code $000, 001, 011, 010, 110, 111, 101, 100$, in which consecutive codes differ in one bit.

5 / COUNTING TOURS

The complete graph has $(n-1)!/2$ Hamilton cycles

Fixing a start and ignoring direction, K_5 has $4!/2 = 12$ cycles, K_{10} has $9!/2 = 181\,440$, and K_{20} has $19!/2 \approx 6.1 \times 10^{16}$, so a salesman visiting 20 towns cannot simply try every route.
The travelling salesman problem asks for the shortest cycle in a weighted complete graph, and finding a Hamilton cycle at all is among the problems with no known fast method.

6 / TRY THESE

Decide and check

(a) Find a Hamilton cycle of K_4: $1, 2, 3, 4$ returning to 1, and there are 3 in all.
(b) Does the star $K_{1,3}$ have one? No, since a leaf has degree 1 and a cycle needs degree 2 at every vertex.
(c) Is a graph with a vertex whose removal disconnects it Hamiltonian? No, a cycle through all vertices would survive the removal of any one vertex.

ASIDE / A HARD PROBLEM

Deciding whether a graph has a Hamilton cycle is NP-complete, which was shown by Karp in 1972.

LINKS TO OTHER SHEETS

The previous sheet gives the Euler test; the weighted networks sheet meets the travelling salesman problem again beside the easy shortest path problem.

Convention. A Hamilton cycle has n edges on n vertices; K_n is the complete graph; Q_3 the cube graph; Dirac and Ore give sufficient conditions.